**Assignment 1**

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## 1. Introduction

In this project, we are adding to project 0 with only allowing edits in the region of interests (ROIs). The program takes a parameter file that includes all the parameters that will be used to do image manipulation and the starting point of x/y and the size of the ROI with respect to x and y. We then were told to add the functions double thresholding on gray level image, color modification, 2d smoothing,and  incremental smoothing

## 2. Description of algorithm

In this section the basic algorithms used in this assignment are described. First the algorithm of the addGrey function. In this section the basic algorithms used in this assignment are described. First the algorithm of increasing the intensity, addGrey function, is described. An algorithm for binarization is then described next. Scaling the image is the third algorithm, and lastly thresholding add is described lastly.

### 2.1. Add grey

In this function we are increasing/decreasing the intensity of a gray-level image with a given value. Are parameters include the intensity we want to increase a pixel to. As discussed in the report 0 we simply iterate through the photo's pixels and change the value of the tgt image which is our output by adding the desired value to it. The changes that were made was to add a ROI. This simple was done by making the function check if a pixel was in the window space of the ROI by using a for loop.

```
tgt.setPixel(i,j,checkValue(src.getPixel(i,j)+value));
```

Figure 1.1 (actual snippet of code that increases the pixel intensity)

### 2.2. Binarize

In this function we binarize the image. This function simply checks if a pixel is less than the given threshold if it is then it is set to the MINRGB of 0. Else, the pixel is set to the MAXRGB of 255. The changes that were made was to add a ROI. This simply was done by making the function check if a pixel was in the window space of the ROI by using a for loop.

### 2.3. Scale

In this function we include a series of mathematical formulas to scale an image to a given ratio provided by parameter text file. Thus, expanding/reducing the image's height and width by 2 or .5. Included is a variable *rows* and *cols* of type **int** that get the number of rows and columns of the input image and are multiplied by the given ratio. The function resize is then called which parameters of that from *rows* and *cols* variables. The function then includes and double for loop similar that starts at *i* = 0 and continues till it is less than *rows*. The inner for loop starts at *j = 0*  and iterates while *j* is less than cols. Inside the inner for loop, there is a series of lines of code that map the pixel of the new image back to the original.

### 2.4. Thresholding add

This function modifies the brightness of a pixel when the intensity is not equal to the threshold. Given a threshold T and two values V1, V2, the intensity is increased with V1 if the intensity of a  pixel in the image is larger than T. The intensity is decreased with V2 if the intensity of a pixel is smaller than T. If a pixel is the same as T then nothing is done to the pixel. The function first resizes the output image *tgt*  with the number of rows and columns of the input image *src*. A for double for loop is then used to change the intensity of a pixel. Starting at index *i = 0* and *j = 0* the first loop iterates while *i* is less than *src's number of columns.* The second loop continues while *j* is less the srcs's number of columns. We now check if a pixel is greater, less than, or equal to the threshold using if states. Once a pixel has entered one of the if/else if statements we first have to check if the RGB range is still from 0-255 for the statements for greater than or less than to after we increase the intensity of the pixel. If the RGB range is still correct we can now set the pixel of the *tgt* image with V1 or V2. Else, we set the pixel to MAXRGB of 255 if the pixel was greater than the threshold and set it to MINRGB of 0 if the pixel was less than the threshold. If the pixel is equal to we set the pixel to that of the *src* image. The changes that were made was to add a ROI. This simple was done by making the function check if a pixel was in the window space of the ROI by using a for loop.

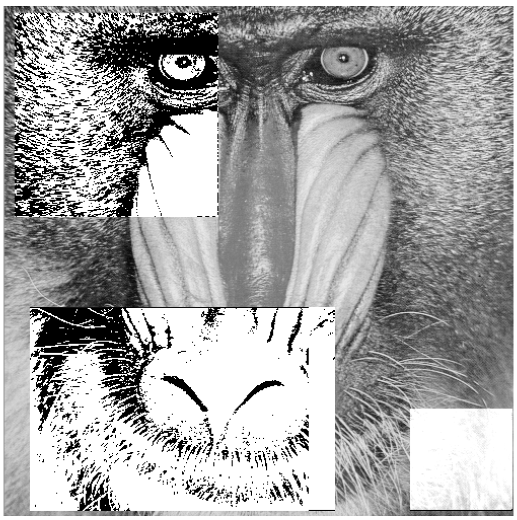
## 3. Description of Implementation

The program was created using C++ a mac os using visual studio code was used to write the code. A sample code was given and what was edited was simply the utility.cpp, utility.h, ip.tool.cpp files to create the function to do thresholding adding.

# 4. Description and Analysis of Results

## 4.1. Description of Results

This section illustrates the results of the algorithm used for Thresholding adding with three ROIs with parameters **threshold1 = 100 threshold2 = 100, threshold3 = 100, v1 = 50, v2 = 30, v1_2 = 50, v2_1 = 30, v1_3 = 50, v2_3 = 30, x1 = 50, y1 = 50, Sx1 = 50, Sy1 = 50, x2 = 200, y2 = 200, Sx2 = 100, Sy2 = 100, x3 = 310, y3 = 310, Sx3 = 100, Sy3 = 10 (NOTE:** v1_2 stand for the first value for the second window, V2_2 would stand for second value for second window and so on).



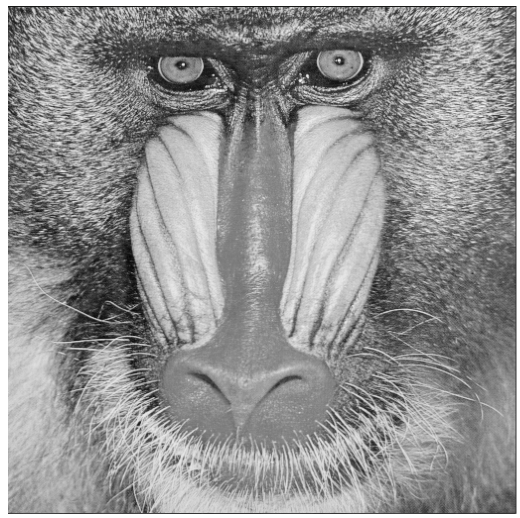


**Figure 1.8 (modify two ROIs with Biniarize and one with addGrey)** **Figure 1.9 (Original photo)**

## 4.2. Performance Evaluation and Analysis of Results

All of the functions run in O(N^2) time due to them all having nested for loops where N is the size of the image. With small images that were tested this code would run “fast” but it will become slower as the size of the image increases. Using different parameters I would see the difference in intensity of the images based on the threshold I used and the input of V1 and V2.

# 5. Conclusion

In this project, we are told to modify project 0 to allow changes only in the ROIs. We also were told to add double thresholding-holding on the grey level image, color modification, 2d smoothing, and sep2dsmooth. The program took a parameter file that includes all the parameters that will be used to do image manipulation. A function to modify the brightness of a pixel was added which algorithm was a combination of the intensity filter and binarize filter with the use of restrictions such as the function would increase the intensity with V1 if the intensity is greater than T and decrease the intensity with V2 if the pixel is smaller than T. If the Intensity of the pixel was equal to the threshold it would stay the same.